

Estimating Interventional Outcomes over Time with Causal Normalizing Flow

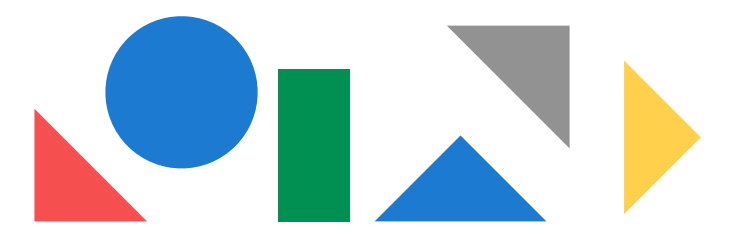
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Motivation

- **Causal inference over time** is central to healthcare and policy: *what happens to future outcomes under a hypothetical treatment strategy?*
- Most time-series causal methods return **point estimates** — they miss the variability that decision-making requires.
- Causal generative models estimate **interventional distributions**, but existing approaches are largely restricted to static settings.

Background

- **Structural causal models (SCMs)** characterize a data-generating process (DGP) through structural assignments with exogenous variables, inducing a causal graph $\mathcal{G}$:

$$X^j = f^j(\mathbf{Pa}^j, Z^j), \quad \text{for } j = 1, \dots, d$$

- **Causal normalizing flows (CNFs; Javaloy et al., 2023)** seek to recover an SCM by fitting an autoregressive flow aligned with $\mathcal{G}$.

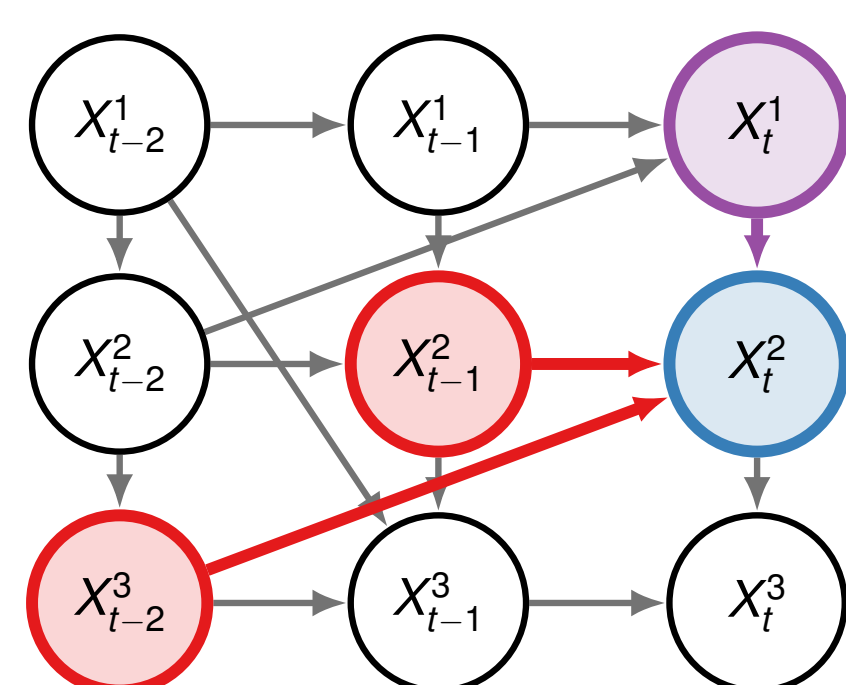
$$x^j = f_{\theta}^j(z^j; \mathbf{Pa}^j), \quad \mathbf{z} \sim P(\mathbf{Z})$$

Problem Setup

For a multivariate time-series $\mathcal{X} \in \mathbb{R}^{T \times d}$, we assume its DGP admits a *Markovian, causally stationary* SCM.

Window causal graph $\mathcal{G}$ — induced by the SCM and assumed given:

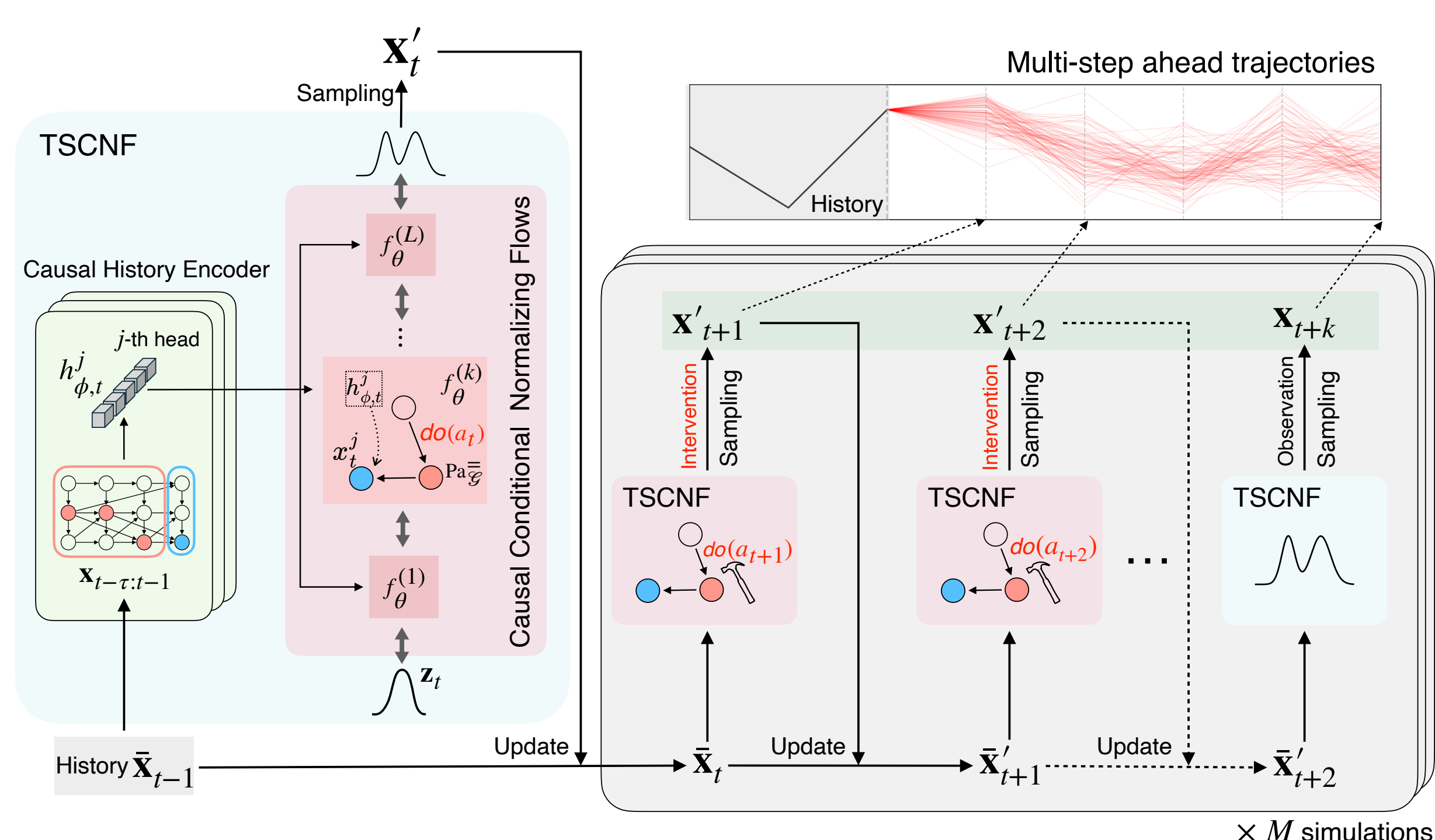
- $\mathbf{Pa}_{\mathcal{G}}^{\leftarrow}(X_t^j)$ — **time-lagged parents**
- $\mathbf{Pa}_{\mathcal{G}}^{\rightarrow}(X_t^j)$ — **contemporaneous parents**



Target estimand — the distribution of the future outcome Y_t under a hypothetical intervention sequence, conditioned on the history:

$$P(Y_t(\underline{a}_{m+1}) \mid \bar{\mathbf{x}}_m), \quad t \geq m+1$$

TSCNF: Time-Series Causal Normalizing Flows



- The joint observational distribution factorizes according to $\mathcal{G}$:

$$P(\mathcal{X}) = \prod_{t=1}^T \prod_{j=1}^d P(X_t^j \mid \mathbf{Pa}_{\mathcal{G}}^{\leftarrow}(X_t^j), \mathbf{Pa}_{\mathcal{G}}^{\rightarrow}(X_t^j))$$

- **Causal History Encoder** — **encodes** $\mathbf{Pa}_{\mathcal{G}}^{\leftarrow}(X_t^j)$ into an embedding $h_{\phi,t}^j$ using feature-wise causal (parent) masking.
- **Causal Conditional Normalizing Flows** — **model** contemporaneous causal dependencies, with each feature conditioned on $h_{\phi,t}^j$.
- **Trained** by maximizing the observational log-likelihood:

$$\max_{\theta, \phi} \sum_{t=1}^T \sum_{j=1}^d \log P_{\theta}(X_t^j \mid \mathbf{Pa}_{\mathcal{G}}^{\leftarrow}(X_t^j), h_{\phi,t}^j)$$

Estimation of Interventional Distribution

- The target estimand admits the following form:

$$P(Y_t(\underline{a}_{m+1}) \mid \bar{\mathbf{x}}_m) = \int_{\mathbf{x}'_{m+1:t-1}} P(Y_t \mid do(\underline{a}_t), \bar{\mathbf{x}}_{t-1}) \prod_{k=m+1}^{t-1} P(\mathbf{x}'_k \mid do(\underline{a}_k), \bar{\mathbf{x}}_{k-1})$$

- Analytically intractable $\Rightarrow$ **Monte Carlo simulation** with TSCNF:

For each time step $k = m+1, \dots, t$:

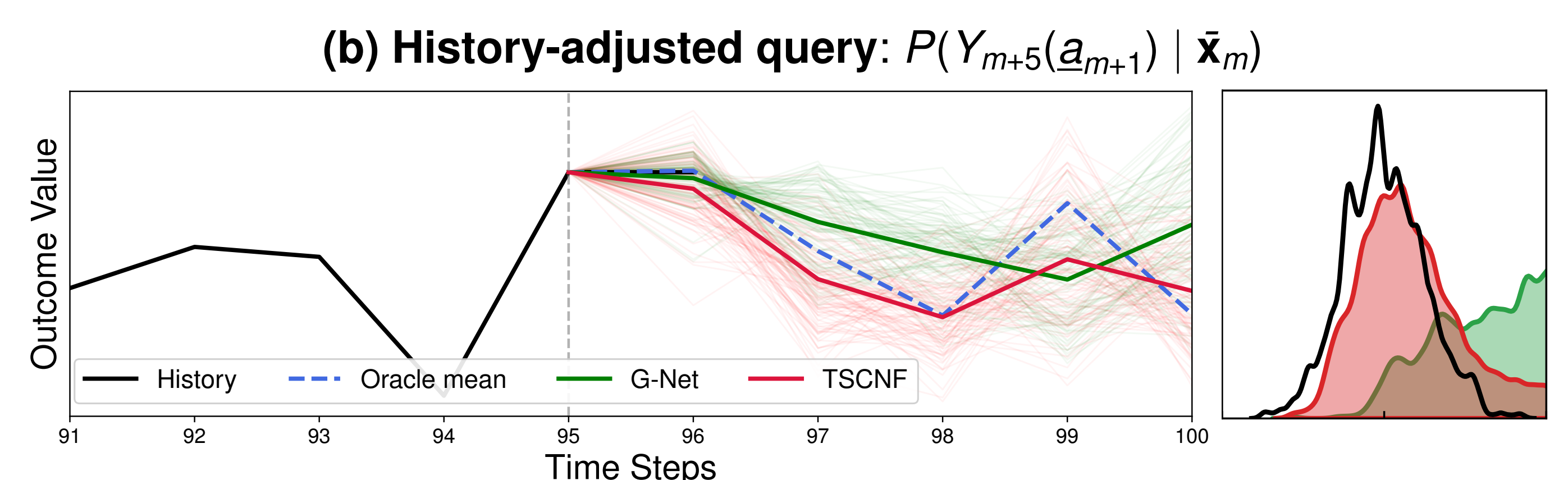
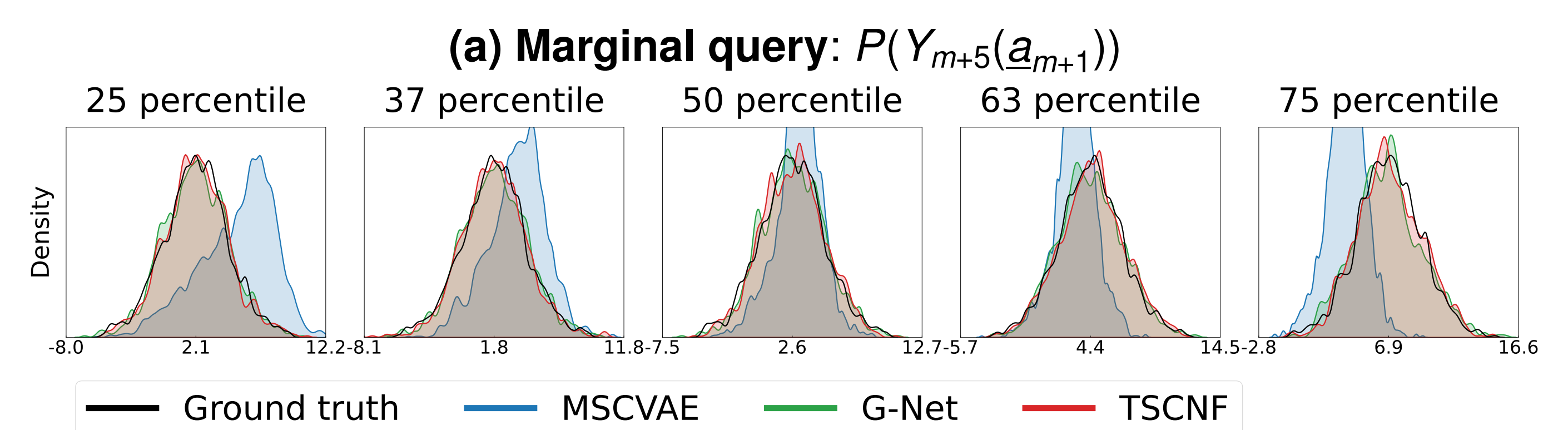
- ① **Encode**: the current history $\bar{\mathbf{x}}_{k-1}$ into $\mathbf{h}_{\phi,k}$.
- ② **Intervene**: fix $A_k = a_k$; sample $\mathbf{x}'_k \sim P(\cdot \mid do(a_k), \bar{\mathbf{x}}_{k-1})$.
 - Infer the latent corresponding to the fixed $A_k = a_k$.
 - Sample the remaining latents $\mathbf{z}'_k \sim P_{\mathbf{z}}$.
 - Push all latents through $f_{\theta}(\cdot \mid \mathbf{h}_{\phi,k})$ to generate $\tilde{\mathbf{x}}_k = [a_k, \mathbf{x}'_k]$.
- ③ **Append**: update the history as $\bar{\mathbf{x}}_k \leftarrow [\bar{\mathbf{x}}_{k-1} \parallel \tilde{\mathbf{x}}_k]$.

At $k = t$, retain Y_t from $\tilde{\mathbf{x}}_t$.

Run M rollouts to form the empirical interventional distribution.

Experiments

A. Synthetic Dataset: Non-linear DGPs



- **Marginal**: TSCNF and G-Net outperform MSCVAE by avoiding unstable propensity weighting.
- **History-adjusted**: Under sparse graphs, TSCNF outperforms G-Net by using only causally relevant signals.

B. Tumor Growth Dataset: PK-PD Simulation (Geng et al., 2017)

nRMSE % (lower is better); best in bold , second best <u>underlined</u>					
Model	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
$\gamma_c = 5, \gamma_d = 5$					
RMSN	0.63 (0.11)	0.74 (0.12)	0.79 (0.11)	0.83 (0.11)	0.88 (0.11)
G-Net	0.52 (0.03)	0.64 (0.04)	0.72 (0.06)	0.79 (0.09)	0.92 (0.24)
Causal CPC	0.70 (0.10)	<u>0.77 (0.10)</u>	0.81 (0.10)	0.84 (0.12)	0.88 (0.14)
DoFlow	1.01 (0.05)	1.54 (0.06)	1.90 (0.06)	2.19 (0.07)	2.40 (0.09)
TSCNF	<u>0.55 (0.09)</u>	0.58 (0.10)	0.63 (0.10)	0.67 (0.10)	0.71 (0.11)
$\gamma_c = 5, \gamma_d = 0$					
RMSN	0.04 (0.02)	0.04 (0.02)	0.04 (0.02)	0.05 (0.02)	0.06 (0.02)
G-Net	0.02 (0.01)	0.04 (0.01)	0.07 (0.09)	0.13 (0.26)	0.19 (0.30)
Causal CPC	0.12 (0.04)	0.12 (0.04)	0.13 (0.05)	0.15 (0.07)	0.18 (0.08)
DoFlow	0.34 (0.04)	0.35 (0.05)	0.38 (0.04)	0.42 (0.05)	0.45 (0.07)
TSCNF	0.02 (0.02)	0.02 (0.01)	0.03 (0.01)	0.03 (0.01)	0.04 (0.02)

- **Best in most settings**—consistently strong across various confounding structures (γ_c, γ_d) .

Contributions

- ① **Extend causal normalizing flows** to the time-series setting.
- ② **Design a learning & inference framework** for multi-step interventional distribution estimation.
- ③ **Generalize across query types** without requiring query-specific design.